

AWRaCLe: All-Weather Image Restoration using Visual In-Context Learning

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- **The Challenge:** All-Weather Image Restoration (AWIR) is difficult because real-world adverse weather (rain, snow, haze) presents highly variable degradations.
- **The Gap:** Prior approaches rely heavily on learning degradation cues from individual training samples but fail to utilize additional contextual information to guide the restoration.
- **The Present Shift:** Visual in-context learning introduces generalist models that use contextual information as a prior.
- **The Proposed Solution:** AWRaCLe is a novel framework that uses degradation-specific visual context (a clean and degraded image pair) to support the restoration network.

Limitations of Existing Frameworks

- **Single-Image Bottleneck:** Previous AWIR models struggle to disentangle scene characteristics from Degradation-Specific Information (DSI) because they rely on extracting features from a single degraded image.
- **Failure of Prior In-Context Models:** Existing visual in-context models (like Painter and PromptGIP) fail for image restoration due to a critical flaw:
 - **Context Suppression:** Context is only injected at the encoder's input. In deep networks, this information is suppressed and becomes negligible by the time it reaches the decoder.

The AWRaCLe Framework

- **The Context Pair:** The model utilizes a context pair $\mathcal{C} = \{I_d, I_c\}$, containing a degraded image I_d and its exact clean counterpart I_c .
- **Scene Consistency:** Because the underlying scene is identical, the primary distinction is strictly the weather degradation.
- **Core Modules:**
 - **DCE (Degradation Context Extraction):** Extracts explicit DSI from the context pair.
 - **CF (Context Fusion):** Integrates this DSI into a modified Restormer restoration network.
- **Multi-Level Fusion:** To solve the context suppression problem, DCE and CF blocks are integrated at every spatial level of the Restormer decoder ($l = 0, 1, 2, 3$).

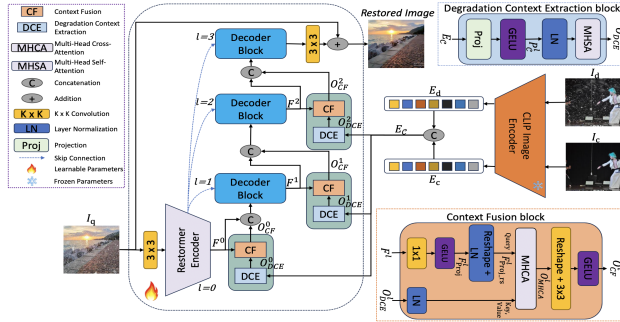


Figure 2: Block diagram of the proposed visual in-context learning approach for AWIR. CLIP features are extracted from I_d and I_c which are subsequently fed to DCE blocks at different decoder levels, l . CF blocks then fuse the degradation information obtained from the DCE blocks with decoder features, F^l , from the query image I_q . Finally, the restored image is generated.

Extracting Degradation-Specific Information (DCE)

- **Feature Extraction:** The process begins by extracting embeddings from the final transformer block of CLIP's image encoder:

$$E_d = \text{CLIP}(I_d), \quad E_c = \text{CLIP}(I_c), \quad \{E_d, E_c\} \in \mathbb{R}^{L \times D}$$

(where L is the number of tokens and D is the embedding dimension)

- **Concatenation & Projection:** Embeddings are concatenated into $E_C \in \mathbb{R}^{2L \times D}$, projected to a lower dimension C' , and activated via GELU:

$$P_C^I = \text{GELU}(\text{Proj}(E_C)), \quad P_C^I \in \mathbb{R}^{2L \times C'}$$

- **Attention Mechanism:** Multi-Head Self-Attention (MHSA) isolates the DSI by directly comparing the clean and degraded tokens:

$$O_{\text{DCE}}^I = \text{MHSA}(\text{LN}(P_C^I)), \quad O_{\text{DCE}}^I \in \mathbb{R}^{2L \times C'}$$

Fusing Context via Cross-Attention (CF)

- **Feature Alignment:** At decoder level l , Restormer features F^l are projected to match the channel dimension C^l of the extracted context:

$$F_{\text{Proj}}^l = \text{GELU}(\text{Conv}_{1 \times 1}(F^l)), \quad F_{\text{Proj}}^l \in \mathbb{R}^{C^l \times H^l \times W^l}$$

- **Dimensionality Matching:** F_{Proj}^l is reshaped (RH) to match the 2D dimensions of O_{DCE}^l :

$$F_{\text{Proj_rs}}^l = \text{LN}(\text{RH}(F_{\text{Proj}}^l)), \quad O_{\text{DCE}}^l = \text{LN}(O_{\text{DCE}}^l)$$

- **Multi-Head Cross-Attention (MHCA):** The decoder features act as the Query, while the extracted weather context acts as Key and Value:

$$F_{\text{Proj_rs}}^l \rightarrow Q, \quad O_{\text{DCE}}^l \rightarrow K, V$$

$$O_{\text{MHCA}}^l = \text{MHCA}(Q, K, V), \quad O_{\text{MHCA}}^l \in \mathbb{R}^{H^l \cdot W^l \times C^l}$$

- **Final Fusion:**

$$O_{\text{CF}}^l = \text{GELU}(\text{Conv}_{3 \times 3}(\text{RH}(O_{\text{MHCA}}^l))), \quad O_{\text{CF}}^l \in \mathbb{R}^{C^l \times H^l \times W^l}$$

Illustration of AWRaCLe (Snow)



Training and Evaluation Datasets

Desnowing:

- **Training:** VOC snow dataset with 900 sample).
- **Evaluation:** VOC snow dataset with 100 samples.

Deraining:

- **Training:** VOC rain dataset with 900 samples.
- **Evaluation:** VOC snow dataset with 100 samples.

Dehazing:

- **Training:** VOC fog dataset with 900 samples.
- **Evaluation:** VOC snow dataset with 100 samples.

Training Details: AdamW optimizer, L_1 loss, random cropping (128×128), and CLIP inputs resized to 224×224 .